KAIROS: Object Discovery from Oscillatory Timing in a Spiking Network

Anonymous submission

Abstract

Can spike timing reveal which patches in a static image belong to the same object? KAIROS models image patches as coupled oscillators whose oscillatory activity controls when patches spike. We hypothesize that synchronization produces similar spike patterns in patches of the same object. KAIROS combines this similarity with image-derived connections to form object masks. It trains without ground-truth masks and infers the number of objects. Under a shared patch-level protocol, KAIROS achieves 0.841 FG-ARI on CLEVR versus 0.633 for the strongest evaluated baseline and scores higher on Tetrominoes and Multi-dSprites without additional training. On 40 held-out CLEVR images, independently shifting each patch's spike trace reduced FG-ARI from 0.865 to 0.074, indicating that our readout relies on the relative timing of activity across patches.

1. Introduction

Object-centric models such as Slot Attention bind distributed image features by assigning them to shared object representations (Locatello et al. 2020). Oscillatory models such as AKOrN add synchronization to feature processing and obtain masks by clustering the resulting representations (Miyato et al. 2025). Both leave open a complementary question: can the timing of neural activity itself tell us which regions belong to the same object?

Neurons signal through brief electrical pulses, and both how often and when they fire can matter: cortical neurons reproduce spike timings under the same fluctuating input (Mainen and Sejnowski 1995). Brain rhythms make firing more likely at some points of a cycle than others, so when two populations synchronize their windows align (Fries 2005); in cat visual cortex the strength of synchrony between sites depended on how their local stimuli formed a larger pattern (Gray et al. 1989). S^2-Net (Dan and Wu 2026), which our core follows, uses coupled oscillators to regulate spiking.

We hypothesize that when oscillators representing patches of the same object synchronize, their downstream neurons produce similar spike patterns.

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2. Method

As shown in Figure 1, we map image patches to coupled oscillators and use their phase dynamics to modulate spiking. We then combine temporal similarity with image-derived connectivity to group patches into object masks.

2.1 From Images to Neural Activations

2.1.1 Patch activation We encode an RGB image with eight convolutional filters, normalize and bound the feature maps, and pool them into a patch activation tensor $\gamma \in \mathbb{R}^{64\times64\times8}$. Each patch i has an eight-dimensional activation vector $\gamma_i \in \mathbb{R}^8$. The patch indices, i and j, identify image patches; the component index, a, identifies an oscillator component; and the time index, t, identifies a simulation step.

2.1.2 Patch connection graph In its neuroimaging setting, S^2-Net (Dan and Wu 2026) uses structural connectivity to set how strongly brain regions interact. We apply the same principle to images: a weighted graph controls how strongly patch oscillators influence one another, with edge weights based on feature similarity and geodesic distance.

$$S_{ij} = \frac{\boldsymbol{z}_i^\top \boldsymbol{z}_j}{\tau} - r\, d(i,j). \tag{1}$$

The connection score, S_{ij}, compares the normalized patch embeddings, $\boldsymbol{z}_i$ and $\boldsymbol{z}_j$, using the temperature, τ, and penalizes the geodesic distance, $d(i,j)$, using the spatial cost weight, r. τ and r are derived from trainable scalars.

We normalize the strongest connections and make their weights reciprocal.

$$\begin{aligned} B_{ij} &= G\ \mathrm{softmax}_{j\in\mathcal{N}_i}(S_{ij}), \\ A_{ij} &= \frac{B_{ij} + B_{ji}}{2}. \end{aligned} \tag{2}$$

For each patch i, the neighbor set, $\mathcal{N}_i$, contains the 512 patches with the highest connection scores, S_{ij}. The connection gain, G, is derived from a trainable scalar. The normalized connection weight, B_{ij}, is zero for unselected neighbors. The symmetric connection weight, A_{ij}, defines the patch graph.

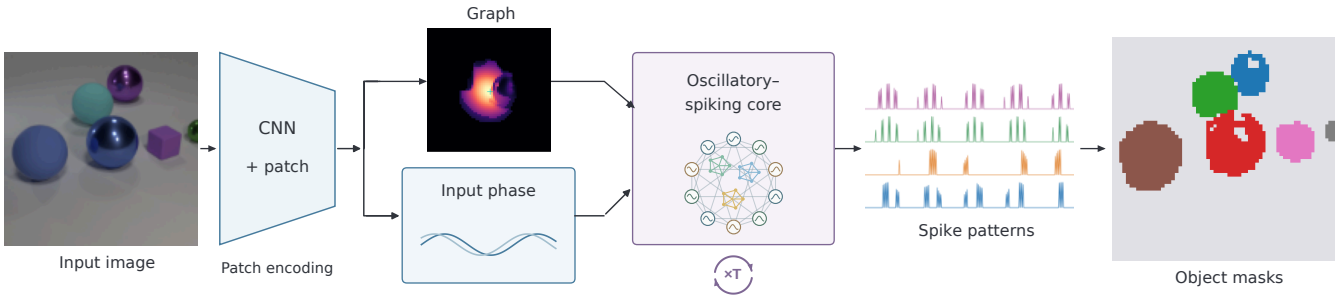


Figure 1: KAIROS overview. CNN patch features define a patch graph and input phases. The oscillatory spiking core runs for T steps, and the readout combines temporal spike similarity with graph connectivity to group patches into object masks.

2.1.3 Input phase We project each patch activation with the trainable weight W_{drive}, standardize the result, and scale it with the trainable gain W_ϕ. A scaled tanh bounds the sensory drive, $\boldsymbol{\phi}_i$, to $[-\pi, \pi]$. We use the bounded drive as the initial oscillator phase (Eq. (A5)).

2.2 Recurrent Core

2.2.1 Kuramoto layer The Kuramoto layer coordinates oscillator phases across patches through intrinsic drift, patch coupling, and sensory drive. Each patch has a phase vector $\boldsymbol{\theta}_i(t) \in \mathbb{R}^4$, with one entry per oscillator.

$$\begin{aligned}\boldsymbol{\theta}_i(t+1) = \boldsymbol{\theta}_i(t) + \Delta t\Big(&\boldsymbol{\omega}_i + f\boldsymbol{\phi}_i \\ &+ \boldsymbol{\kappa}_i \odot \sin(\boldsymbol{\phi}_i - \boldsymbol{\theta}_i(t)) \\ &+ \frac{K}{N}\sum_j \bar{A}_{ij}\sin\big(\boldsymbol{\theta}_j(t) - \boldsymbol{\theta}_i(t) - \alpha_{ij}\big)\Big).\end{aligned} \tag{3}$$

The **natural frequency**, $\boldsymbol{\omega}_i$, gives each oscillator its intrinsic rate of phase change. The **coupling term**, the sum over j, adjusts the phase according to other patches, using the corrected connection weight, $\bar{A}_{ij}$, and the phase lag, α_{ij}. The **drive term** uses the sensory drive, $\boldsymbol{\phi}_i$, to adjust frequency through f and to pull the phase toward the input when the drive strength, $\boldsymbol{\kappa}_i$, is positive. The coupling scale, K, is normalized by the patch count, N. The time step, Δt, scales the update, and $\odot$ denotes elementwise multiplication. The parameters $\boldsymbol{\omega}_i$, f, and $\boldsymbol{\kappa}_i$ are trainable.

2.2.2 Sinusoidal gating The sinusoidal gating layer modulates the signal sent to the dendrites using a delayed phase. Each patch has one scalar gate $g_i(t) \in [0, 1]$, shared by its four oscillator components.

$$g_i(t) = \frac{1 + \sin(\bar{\theta}_i(\max(0, t-d)))}{2}. \tag{4}$$

$$\boldsymbol{x}_i(t) = g_i(t)\sin(\boldsymbol{\theta}_i(t)). \tag{5}$$

The gate $g_i(t)$ uses the mean phase of the four oscillators (Eq. (A8)) with a phase delay of d steps to modulate the dendritic input $\boldsymbol{x}_i(t)$.

2.2.3 From Gated Activity to Spikes Four dendritic branches integrate the current gated input with top-down feedback from the previous spike. Each branch balances its previous state with an update computed from these inputs, and the oscillatory gate modulates how much the state changes. The branch outputs are summed to drive the membrane. The membrane similarly balances its previous voltage with an update driven by the current dendritic input, with the gate modulating the size of the voltage change. When the voltage exceeds a threshold, the membrane generates a spike event, and the gate scales the emitted output. Appendix B provides the full update equations.

2.3 Spike Grouping and Mask Generation

For each oscillator component, we mean-center and L2-normalize the spike trace from the last 512 simulation time steps. For patches i and j, the correlation ρ_{ija} is higher when their spike activity tends to be above or below its respective average at the same time (Eq. (A16)). We compute the patch affinity, Q_{ij}, by multiplying the four correlations after clipping negative values to zero. A high affinity thus requires agreement across all four components: any nonpositive correlation makes Q_{ij} zero.

We form connected components by linking patch pairs with $Q_{ij} \geq 0.5$ and discard components smaller than 8 patches, a floor selected on CLEVR validation. Every patch is a foreground candidate; no firing-rate test is applied. As a heuristic, we treat the largest remaining component as background, and the components that survive form the foreground support.

We start the retained foreground patches as singleton groups and merge spatially adjacent groups using

$$\begin{aligned}E_{ij} &= \left(1 + \frac{N\bar{A}_{ij}}{G}\right)\log\frac{Q'_{ij}}{1 - Q'_{ij}}, \\ \Delta(\mathcal{C}, \mathcal{D}) &= \sum_{i\in\mathcal{C}}\sum_{j\in\mathcal{D}} E_{ij}.\end{aligned} \tag{6}$$

The stabilized affinity, Q'_{ij}, determines the sign of the merge evidence, E_{ij}. The merge score, $\Delta(\mathcal{C}, \mathcal{D})$, sums the evidence between the candidate patch groups, $\mathcal{C}$ and $\mathcal{D}$. Groups are merged while their score is positive. Each final group produces an object mask.

Model	CLEVR (in-domain)			Tetrominoes (zero-shot)			Multi-dSprites (zero-shot)		
	ari	f-iou	o-iou	ari	f-iou	o-iou	ari	f-iou	o-iou
KAIROS (ours)	**.8408**	**.7528**	**.6387**	**.8568**	**.7950**	**.8170**	**.6966**	**.8190**	**.7625**
AKOrN (attn) + up-tiling	.6326	.3468	.3636	.5484	.7230	.4167	.5505	.6539	.6348
AKOrN (conv)	.6046	.4219	.2769	.7202	.7540	.5913	.4055	.6834	.5434
AKOrN (attn)	.6027	.3327	.3750	.5676	.7383	.4687	.4343	.6019	.5751
ItrSA	.5794	.2787	.3853	.6319	.7137	.5687	.4817	.2489	.6093
CAE++	.2838	.2193	.0882	.2715	.2452	.1901	.2648	.1048	.1125
Slot Attention	.2872	.2220	.0801	.1739	.2192	.1071	.1930	.0876	.0680
CtCAE	.2357	.2148	.0964	.2681	.2420	.2034	.3079	.1158	.1609
CAE	.1091	.2008	.0634	.1526	.2305	.1192	.1200	.1038	.0630
Best baseline, per cell	.6326	.4267	.3853	.7202	.7540	.5913	.5505	.6834	.6348
Our margin	**+.2082**	**+.3261**	**+.2534**	**+.1366**	**+.0410**	**+.2257**	**+.1461**	**+.1356**	**+.1277**

Table 1: All models trained on the same 10,000 CLEVR images for 10 epochs and evaluated on 320 held-out images per dataset by the same function; Tetrominoes and Multi-dSprites are zero-shot. Metrics are foreground ARI, foreground IoU, and matched-object IoU on the 64×64 patch grid. Every model uses the minimum-region-size rule of Section 3.1 at $f = 0.01$. The penultimate row takes the best baseline in each cell separately, which no single baseline attains. We report one training seed. The baselines' cluster count and background mapping are selected on each target dataset's own validation split, and ours on CLEVR validation only. Appendix E states each model's conditions.

3. Experiments

3.1 Protocol

We evaluate all models with the same scoring function on a 64×64 patch grid. We quantize ground-truth masks to this grid, repeat labels exactly for coarser predictions, and use patch majority voting for finer predictions. We never upsample predictions to pixels for scoring. This protocol has a score ceiling of 1.0, and its scores are not directly comparable to published pixel-level scores.

We report foreground adjusted Rand index (FG-ARI), foreground IoU (FG-IoU), and matched-object IoU (Obj-IoU) separately. Background has label 0 in both maps. FG-ARI compares partitions only on ground-truth foreground patches, so false foreground on true background does not affect it. FG-IoU compares the unions of predicted and true foreground and therefore penalizes false foreground. Appendix E gives the metric definitions.

Training exposure. We train every model on the same 10,000 CLEVR images for 10 epochs. For KAIROS, this corresponds to 6,250 updates with a batch size of 16, or 100,000 image exposures. We adjust only the epoch count in published baseline recipes and retain their architectures and optimizers. Table 1 evaluates transfer to Tetrominoes and Multi-dSprites without training on either dataset. Appendix E distinguishes training from validation-based tuning on the target dataset.

Baseline readouts. Several baselines obtain masks by clustering continuous embeddings. The reference implementations of CAE, CAE++, CtCAE, and SynCx set the cluster count to the number of unique ground-truth labels in each image. We instead use a fixed cluster count per dataset and select it and the background mapping on that dataset's validation split. This gives those baselines access to target validation data that KAIROS does not use for transfer. We make no choices on the reported split.

Minimum region size. Single-patch regions account for 22.8% of KAIROS predictions on CLEVR validation, 27.0% on Tetrominoes, and 56.2% on Multi-dSprites. These regions arise when greedy merging splits a scattered synchronous component. They increase the predicted count with little effect on the partition. We discard regions smaller than a fraction f of the image's predicted foreground. The rule uses no ground-truth labels at inference. We set $f = 0.01$ on CLEVR validation and apply it unchanged to both transfer datasets and all baselines.

3.2 Generalization across datasets

With matched training exposure and one training seed, KAIROS scores higher on all nine dataset–metric combinations. The reference row in Table 1 reports the best baseline for each cell, drawing from three architectures. For example, AKOrN (conv) is the closest baseline on Tetrominoes, AKOrN (attn) with up-tiling on Multi-dSprites, and ItrSA for CLEVR Obj-IoU. The smallest margin is 0.0410 for Tetrominoes FG-IoU; the other margins range from 0.128 to 0.326.

Predicted foreground. KAIROS predicts foreground fractions of 0.1988 on CLEVR and 0.0827 on Multi-dSprites, compared with ground-truth fractions of 0.2115 and 0.0943. On CLEVR, all baselines predict larger foreground fractions than the ground truth, ranging from 0.41 to 0.86.

Training budget and released weights. Slot Attention's released checkpoint, trained with roughly 320 times our exposure budget, scores lower in all nine cells by 0.377 to 0.720. Two of six baselines reach their published scores at 1/210 and 1/64 of their original schedules, respectively. For the other four, the results characterize performance at the matched training budget. Appendix F reports these comparisons, rollout lengths, and data scaling; Appendix G gives per-model results.

3.3 Timing and object grouping

We use the same rollouts from the first 40 held-out CLEVR images for three analyses of the hypothesis in Section 1.

Affinity within and across objects. We use ground-truth labels only to categorize patch pairs, without setting thresholds or object counts. The mean affinity Q is 0.891 ± 0.111 within objects, 0.047 ± 0.048 across objects, and 0.099 ± 0.106 between objects and background. Within-object affinity is about 19 times the across-object affinity.

Temporal alignment. We circularly shift each patch's four spike traces by a patch-specific offset within the readout window. This preserves each patch's firing rate and waveform while disrupting timing across patches. We keep the readout rules fixed and recompute affinities, background selection, and merges. FG-ARI falls from 0.865 to 0.074 and Obj-IoU from 0.693 to 0.033, while the mean region count rises from 7.8 to 562. FG-IoU falls from 0.803 to 0.736 because splitting a region can leave the union of predicted foreground patches nearly unchanged.

Source of the timing signal. The emitted spike is the product of the oscillatory gate and the binary firing indicator. The indicator is 1 in 99.99% of settled steps, and the emitted trace correlates 0.9997 with the gate. Feeding the gate directly into the same readout reproduces every mask on all 40 images. The grouping signal therefore comes from the timing of the oscillatory gate in this experiment; we find no separate contribution from the firing threshold.

3.4 Limitations

Object count. With the minimum-size rule, KAIROS predicts 5.15 objects per image on CLEVR, 2.64 on Multi-dSprites, and 5.28 on Tetrominoes. The true means are 6.36, 2.49, and 3, and exact-count rates are 25.0%, 53.1%, and 9.4%, respectively. Without the rule, predicted means rise to 8.04, 6.62, and 21.07, indicating that single-patch fragments account for much of the count error. Selecting f on target validation data reduces the remaining error: on Tetrominoes, $f = 0.10$ gives a mean of 3.02 and an exact-count rate of 86.9%. We report the transferred setting because target-specific selection uses target labels.

Tetrominoes always contains three objects. Fixed-count baselines are therefore 100% exact even when grouping is poor (CtCAE reaches 0.0756 FG-ARI in this count comparison). On Multi-dSprites, where counts vary, these baselines are 0% exact versus 53.1% for KAIROS.

KAIROS trains without the reconstruction objective used by every baseline in Table 1. The absence of this penalty may help explain why KAIROS still splits some objects into parts.

Spiking nonlinearity. The firing indicator is saturated over the readout window (Section 3.3). The results support a role for oscillatory timing but do not isolate a contribution from thresholding. Whether sparser firing adds grouping information remains open, since the training objective does not constrain the firing rate.

Training seeds. We report one training seed. Reseeding one baseline changes its CLEVR FG-ARI by up to 0.134, less than the CLEVR margins in Table 1. This measures variability for that baseline and does not estimate uncertainty for KAIROS.

Evaluation split. The held-out images come from the training corpus, rather than an independent benchmark test set. We use published scores only for comparisons of each model with its own published results (Appendix F); Table 1 reports scores under our evaluation protocol.

4. Conclusion

KAIROS shows that an oscillatory spiking network can group image patches into object masks without mask supervision during training or a prespecified object count at inference. Under a shared evaluation protocol it reached 0.841 FG-ARI on CLEVR against 0.633 for the best evaluated baseline, and scored higher on Tetrominoes and Multi-dSprites without additional training, at a single training seed. Ablations place the grouping signal in the cross-patch alignment of the oscillatory gate rather than in the firing threshold, so the result supports relative oscillatory timing as a cue for object discovery. The rollout is costly, counting is weaker than grouping, and whether the gated traces also support object tracking is left to future work.

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Appendix

A Sensory Representation and Graph Details

The input is a 128×128 RGB image scaled by $\frac{1}{255}$. A single 3×3 convolution with eight trainable filters, unit stride, and no padding produces $126 \times 126 \times 8$ feature maps. There is no post-convolution activation. All eight channels share a scalar mean μ and standard deviation σ, estimated once from the initial encoder on training RGB images. We use $\sigma = \sqrt{\max(\mathrm{Var}(F), 10^{-8})}$, where F pools feature values over training images, channels, and spatial positions. We standardize with these fixed statistics and clip the result to

$[-3, 3]$. Adaptive average pooling produces a 64×64 grid, yielding $N = 4096$ patch vectors with eight features each.

The bias-free trainable projection W gives the embedding

$$\boldsymbol{z}_i = \frac{W\gamma_i}{\max(\|W\gamma_i\|_2, \varepsilon)}. \tag{A1}$$

For grid coordinates p_i, the scaled spatial distance and local traversal cost are

$$\begin{aligned} d_{\mathrm{sp}}(i,j) &= \frac{\|p_i - p_j\|_2}{4}, \\ c_{ij} &= d_{\mathrm{sp}}(i,j)[1 + 8(1 - \boldsymbol{z}_i^\top \boldsymbol{z}_j)^2]. \end{aligned} \tag{A2}$$

Set $d_{ij}^{(0)} = c_{ij}$ when $d_{\mathrm{sp}}(i,j) \le 1.5$ and 16 otherwise. With $\tau_{\mathrm{geo}} = 0.5$, three updates ($m = 0, 1, 2$) give

$$\begin{aligned} u_{ij}^{(m)} &= -\tau_{\mathrm{geo}} \log \sum_k \exp\left(-\frac{d_{ik}^{(m)} + d_{kj}^{(m)}}{\tau_{\mathrm{geo}}}\right) \\ &\quad + \tau_{\mathrm{geo}} \log 4, \\ d_{ij}^{(m+1)} &= \min(d_{ij}^{(m)}, u_{ij}^{(m)}), \\ d(i,j) &= \min(d_{ij}^{(3)}, 16). \end{aligned} \tag{A3}$$

With learned scalars $\ell_\tau, \ell_G, \ell_r$, the graph parameters are

$$\begin{aligned} \tau &= \max(\exp(\ell_\tau), 0.001), \qquad G = \exp(\ell_G), \\ r &= \mathrm{softplus}(\ell_r) = \log(1 + \exp(\ell_r)). \end{aligned} \tag{A4}$$

Each patch selects its 512 highest-scoring connections before normalization. Symmetrization can increase its degree beyond 512.

The trainable projection, W_{drive}, maps eight features to four components without bias. Its output is standardized over all $4N$ patch-component entries of the current image with a standard deviation floor of 10^{-6}. The trainable scalar, W_ϕ, scales this input before tanh bounds the drive between $-\pi$ and π. The drive and graph remain fixed during a rollout. The input phase is computed as

$$\begin{aligned} \boldsymbol{u}_i &= W_{\mathrm{drive}}\gamma_i, \\ \boldsymbol{\phi}_i &= \pi \tanh(W_\phi \bar{\boldsymbol{u}}_i), \qquad \boldsymbol{\theta}_i(0) = \boldsymbol{\phi}_i. \end{aligned} \tag{A5}$$

B Recurrent Dynamics

B1 Coupling Correction and Phase Lag The corrected graph stabilizes inverse connection costs and gives zero graph entries a small residual weight:

$$\begin{aligned} \bar{A} &= \mathrm{ReLU}\left(\frac{A + A^\top}{2}\right) + 10^{-6}, \\ C_{ij} &= \frac{1}{\bar{A}_{ij}}. \end{aligned} \tag{A6}$$

$$\begin{aligned} \bar{C}_{ij} &= \frac{C_{ij} - \min C}{\max C - \min C + 10^{-6}}, \\ \alpha_{ij} &= \tanh(R_{ij} - R_{ji})\bar{C}_{ij}. \end{aligned} \tag{A7}$$

The inverse connection cost, C_{ij}, is normalized over all patch pairs to obtain $\bar{C}_{ij}$. The trainable directional matrix, R, determines the direction of the phase lag, α_{ij}, through its antisymmetric difference. The inverse cost sets the lag scale. This permits coordinated phases with nonzero offsets. The lag is shared by all four components.

The natural frequency, $\boldsymbol{\omega}_i$, controls intrinsic angular drift. The drive adjusts frequency through $f\boldsymbol{\phi}_i$ and provides sine forcing toward $\boldsymbol{\phi}_i$ when the drive strength, $\boldsymbol{\kappa}_i$, is positive. Coupling, drive, and drift act together; phase is not required to converge to the input phase. The coupling scale, K, equals N, and the time step, Δt, is 0.1.

B2 Rhythmic Gating and Branch Integration We set the phase delay to $d = 2$ simulation steps. The scalar gate, $g_i(t)$, is shared across the patch's four components and has no trainable parameters.

The mean patch phase averages the four component phases:

$$\bar{\theta}_i(t) = \frac{1}{4}\sum_{a=1}^{4} \theta_{ia}(t). \tag{A8}$$

The branch input combines the gated feature, $x_{ia}(t)$, with the previous emitted spike, $s_{ia}(t-1)$, through the trainable projection, $U \in \mathbb{R}^{4\times 2}$, and the trainable bias, $\boldsymbol{b} \in \mathbb{R}^4$:

$$\boldsymbol{v}_{ia}(t) = U\begin{pmatrix} x_{ia}(t) \\ s_{ia}(t-1) \end{pmatrix} + \boldsymbol{b}. \tag{A9}$$

The previous spike is the top-down feedback from the spiking output to the dendritic branches.

Each branch state vector has four entries. The retention vector, $\boldsymbol{\beta}_i$, is obtained by applying sigmoid to the trainable retention vector, $\boldsymbol{k}_i$. The provisional branch state is

$$\begin{aligned} \widehat{\boldsymbol{h}}_{ia}(t) &= \boldsymbol{\beta}_i \odot \boldsymbol{h}_{ia}(t-1) \\ &\quad + (1 - \boldsymbol{\beta}_i) \odot \boldsymbol{v}_{ia}(t). \end{aligned} \tag{A10}$$

The gate and the trainable modulation weight, η_a, set how much of the increment toward the provisional state is applied, and summing the four entries gives the membrane input, $H_{ia}(t)$, where b indexes an entry of $\boldsymbol{h}_{ia}(t)$:

$$\begin{aligned} \boldsymbol{h}_{ia}(t) &= \boldsymbol{h}_{ia}(t-1) + \big(1 - \eta_a(1 - g_i(t))\big) \\ &\quad \cdot \big(\widehat{\boldsymbol{h}}_{ia}(t) - \boldsymbol{h}_{ia}(t-1)\big), \\ H_{ia}(t) &= \sum_{b=1}^{4} h_{iab}(t). \end{aligned} \tag{A11}$$

Retention is shared across oscillator components of the same patch, while their branch states remain separate. Modulating the full increment also controls leakage and spike recurrence, which would continue if only the current input were gated. Summing the four entries combines branch responses without averaging oscillator components.

B3 Membrane Update and Spike Emission The provisional membrane voltage is

$$\begin{aligned} \widehat{V}_{ia}(t) &= \lambda_i V_{ia}(t-1) + (1 - \lambda_i) H_{ia}(t) \\ &\quad - \vartheta_a s_{ia}(t-1). \end{aligned} \tag{A12}$$

The membrane retention, λ_i, is the sigmoid of the trainable patch parameter, ℓ_i. The firing threshold, ϑ_a, is 0.06 times the exponential of the trainable component parameter, q_a.

Membrane resistance is fixed to one. The subtractive reset is proportional to the previous emitted spike and does not force voltage to zero.

As in the branch update, the gate and the trainable modulation weight, ξ_a, set how much of the increment toward the provisional voltage is applied, and the thresholded voltage is gated to give the emitted spike:

$$V_{ia}(t) = V_{ia}(t-1) + \big(1 - \xi_a(1 - g_i(t))\big) \cdot \big(\widehat{V}_{ia}(t) - V_{ia}(t-1)\big). \tag{A13}$$

$$e_{ia}(t) = \mathrm{ind}(V_{ia}(t) > \vartheta_a), \quad s_{ia}(t) = g_i(t) e_{ia}(t), \tag{A14}$$

where ind is the indicator function and $e_{ia}(t)$ is the binary firing event.

When the gate is exactly zero, the previous voltage is retained. Otherwise, the gated update is applied before thresholding. The emitted spike, $s_{ia}(t)$, enters the next branch input and membrane reset. Dendritic and membrane states and the previous spike start at zero. The rollout uses 1024 steps.

C Temporal Affinity and Object Masks

We retain steps $\mathcal{T} = \{512, \ldots, 1023\}$ and index them by $m = 1, \ldots, 512$. Centering and L2 normalization give

$$\mu_{ia} = \frac{1}{512} \sum_{t \in \mathcal{T}} s_{ia}(t), \quad r_{ia,m} = \frac{s_{ia}(511+m) - \mu_{ia}}{\max\big(\sqrt{\sum_{t \in \mathcal{T}} (s_{ia}(t) - \mu_{ia})^2}, 10^{-8}\big)}. \tag{A15}$$

A constant trace becomes zero. The component correlation and patch affinity are

$$\rho_{ija} = \mathrm{clip}\left(\sum_{m=1}^{512} r_{ia,m}\, r_{ja,m}, -1, 1\right), \tag{A16}$$

$$Q_{ij} = \prod_{a=1}^{4} \max(\rho_{ija}, 0). \tag{A17}$$

Centering removes the baseline and normalization reduces sensitivity to absolute scale. The product requires positive temporal agreement in every component; one nonpositive correlation makes the affinity zero. The affinity is not a calibrated same-object probability.

Connected components of $Q \geq 0.5$ determine the initial support without a spatial adjacency constraint. Components smaller than 8 patches are discarded, a floor selected on CLEVR validation, and the largest remaining component is treated as background. Every patch enters this step as a candidate; no activity or firing-rate threshold is applied, so the foreground is defined entirely by the synchrony structure. Removed patches are not recovered. The retained support is reset to singleton patches before merging.

The stabilized affinity is

$$Q' = \mathrm{clip}\left(\frac{Q + Q^\top}{2}, 10^{-6}, 1 - 10^{-6}\right). \tag{A18}$$

The merge evidence, E, is symmetrized and its diagonal set to zero. The log-odds term gives positive evidence above 0.5 and negative evidence below 0.5. Graph weights change its magnitude without changing its sign. Merge scores sum all cross-group pairs. Only groups adjacent on the four-neighbor grid are candidates. The largest positive candidate is merged until none remains.

The hard label, y_i, identifies the final foreground group or equals zero for excluded patches. The binary object mask, M_k, selects the patches assigned to group k on the 64×64 grid. The 8-patch floor is not reapplied after merging. If no foreground survives, the output is all background.

D Training Details

D1 Initialization and optimization We train the Native64 encoder and recurrent core jointly from random initialization without loading a pretrained checkpoint. The shared dendritic bias starts at zero. Training uses RGB images without segmentation masks, object counts, or reconstruction targets.

We compute the objective separately for each image and average the resulting losses across images. After accumulating gradients, we clip the global ℓ_2 norm across trainable parameters to 1 and apply Adam. The encoder and the other active parameter groups use separate fixed learning rates. The legacy geodesic contrast parameter is excluded from optimization because the forward operator uses a fixed contrast.

D2 Rollout and gradient window We simulate 1,024 steps per image. Both losses use steps 512–1023 (zero-based), but recurrent backpropagation covers only steps 960–1023. The first 960 steps run without gradient tracking, and the recurrent states are detached at the boundary. The earlier steps in the loss window still contribute to the statistics. The image graph, sensory drive, and factorized coupling are computed with gradients enabled and reused in the final 64 steps, so the loss can reach the encoder through the graph and drive. The phase loss propagates through the Kuramoto dynamics, while the spike loss also trains the dendritic and membrane parameters through the emitted spike traces. A multi-Gaussian surrogate derivative supports backpropagation through the binary spike threshold. Connected-component filtering and greedy merging are used for inference and do not enter the training objective.

D3 Phase and spike statistics The phase loss uses a phase-locking value (PLV) matrix P. Using the arithmetic mean phase $\bar{\theta}_i(t)$ from Eq. (A8), define $\delta_{ij}(t) = \bar{\theta}_i(t) - \bar{\theta}_j(t)$ and let $\langle \cdot \rangle_t$ denote the mean over steps 512–1023. Then

$$R^{\mathrm{plv}}_{ij} = \langle \cos \delta_{ij}(t) \rangle_t, \quad I^{\mathrm{plv}}_{ij} = \langle \sin \delta_{ij}(t) \rangle_t, \quad P_{ij} = \mathrm{clip}\left(\sqrt{(R^{\mathrm{plv}}_{ij})^2 + (I^{\mathrm{plv}}_{ij})^2 + 10^{-12}}, 0, 1\right). \tag{A19}$$

A high PLV reflects a stable phase difference, which need not be zero. The spike loss uses the affinity matrix Q from Appendix C, computed from the emitted, gated spikes over the same window. Unlike P, Q multiplies the nonnegative correlations of four separate spike traces.

D4 Loss function For $X \in \{P, Q\}$, let $\langle\cdot\rangle_{\neq}$ be the mean over all $N(N-1)$ ordered off-diagonal pairs. We use the same criterion for both matrices:

$$\begin{aligned} \mathcal{C}(X) &= 6L_{\text{bim}}(X) + 10L_{\text{bal}}(X) \\ &\quad + 0.5L_{\text{coh}}(X) + L_{\text{col}}(X), \\ L &= \mathcal{C}(P) + 5\mathcal{C}(Q). \end{aligned} \tag{A20}$$

Let $\mu_X = \langle X_{ij}\rangle_{\neq}$ and $v_X = \langle (X_{ij} - \mu_X)^2\rangle_{\neq}$. The bimodality, balance, and collapse terms are

$$\begin{aligned} L_{\text{bim}}(X) &= \langle X_{ij}(1 - X_{ij})\rangle_{\neq}, \\ L_{\text{bal}}(X) &= (\mu_X - 0.867)^2, \\ L_{\text{col}}(X) &= -\log(v_X + 10^{-4}). \end{aligned} \tag{A21}$$

The first favors affinities near zero or one; the second targets a fixed mean affinity of 0.867; the third penalizes low off-diagonal variance. The target is a pairwise affinity prior, not a target for object count or foreground area. The logarithmic penalty is finite at zero variance.

For spatial coherence, we reshape the off-diagonal row means $a_i = (N-1)^{-1}\sum_{j\neq i} X_{ij}$ into a 64×64 field $a_{r,c}$ and use

$$\begin{aligned} L_{\text{coh}}(X) = {} & \text{mean}_{r,c}\,|a_{r+1,c} - a_{r,c}| \\ & + \text{mean}_{r,c}\,|a_{r,c+1} - a_{r,c}|. \end{aligned} \tag{A22}$$

Each mean includes only valid adjacent pairs in its direction. This term smooths each patch's mean affinity to the other patches; it does not operate on the final masks.

E Evaluation Conditions

E1 What each metric can and cannot see All three metrics take two aligned integer label maps on the 64×64 patch grid, with background as label 0 on both sides. They differ in which patches they look at, and the difference decides which errors are charged.

Foreground ARI restricts both label maps to the patches whose *ground truth* is foreground, and computes the adjusted Rand index of the two partitions of that set. Predicted background is kept as an ordinary cluster inside it, so predicting background on a true object is charged; but patches whose ground truth is background are removed before scoring, so *labelling background as object is invisible to this metric*. An image with fewer than two true foreground patches is undefined and excluded; a degenerate pair of identical partitions scores 1.

Foreground IoU merges every non-background identity on each side into one set and reports $|\hat{F} \cap F|/|\hat{F} \cup F|$ over patches. False foreground enters the union and nothing else, so this is the metric that charges for claiming background as object — the error FG-ARI cannot see. Both sets empty scores 1; exactly one empty scores 0. This is why Table 1 is read with the two together: a method that labels most of the image as object can hold a respectable FG-ARI while its foreground IoU collapses, which is what the F-IOU column shows for the baselines that predict 0.41 to 0.86 foreground against a truth of 0.2115.

Matched-object IoU solves a one-to-one Hungarian assignment between predicted and true objects maximising total IoU, sums the matched IoUs and divides by the number of *true* objects, so an unmatched true object contributes zero. Background is not matched. False foreground enlarges the unions of the objects it touches, but *extra predicted objects carry no direct penalty* beyond going unmatched. Over-segmentation is therefore not charged by this metric, which is the reason object count is reported separately in Section 3.3 rather than treated as covered by it. No true objects gives an undefined score; predictions empty against a non-empty truth gives 0.

Undefined images are excluded per metric and the remainder averaged, so the three metrics can rest on slightly different image counts; the counts are recorded with every result. No weighted combination of the three is formed anywhere.

E2 Which split fixed each free choice Every method has choices that are not learned by its training objective, and a transfer claim is only as strong as the data used to fix them. *Not trained on the target* and *not tuned on the target* are different statements, and they do not hold equally for every row of Table 1.

Ours. One free scalar: the minimum-region-size fraction f. It is fixed at 0.01 on the *CLEVR* validation split and applied unchanged to Tetrominoes and Multi-dSprites. No Tetrominoes or Multi-dSprites data, labelled or unlabelled, enters any choice we make. Everything else in the readout — the $Q \geq 0.5$ link threshold, the 8-patch component floor (itself selected on CLEVR validation), the largest-component background heuristic, the rollout length and the readout window — is fixed before these datasets are touched and is identical across all three.

Baselines. Two free choices, and *both are fixed on the target dataset's own validation split*: the number of clusters k and the rule mapping one cluster to background. The background rule is selected from four candidates (global-largest, image-largest, border-largest, border-union) by validation FG-ARI with foreground IoU as tie-break, separately for each model and each dataset. For the CAE family k is likewise selected per dataset from $\{4, 7, 11\}$, and the selection does not land on the same value everywhere: CAE++ uses $k = 11$ on CLEVR and Tetrominoes but $k = 7$ on Multi-dSprites, and CAE uses 11, 11 and 4. For AKOrN, ItrSA and Slot Attention the slot count is the architecture's own and is left at the authors' value, but the background rule is still chosen on the target's validation split. The selection used 320 validation images on CLEVR, and 1,000 on Tetrominoes and Multi-dSprites for the AKOrN family, 320 for the CAE family.

The asymmetry runs in the baselines' favour, not ours: on the two transfer datasets they are allowed to see target labels to set k and the background mapping, and we are not. We state it here rather than leaving "zero-shot" to carry both meanings at once. Removing the oracle that the reference implementations of CAE, CAE++, CtCAE and SynCx use — setting k_i from the per-image ground-truth object count — is a separate and stricter correction, described in Section 3.1; a per-dataset constant chosen on validation is what replaces it.

The minimum-region-size rule is offered to every model

T	FG-ARI	FG-IOU	OBJ-IOU	Δ FG-ARI
1024	.8421	.7545	.6410	—
512	.8402	**.7594**	.6340	$-.0019$
256	.8159	.7358	.6084	$-.0262$
128	.5796	.4988	.3985	$-.2625$

Table 2: CLEVR, model fixed, readout window $T/2$. Halving the rollout is free; a quarter costs 0.02–0.03; an eighth fails. This table and the scaling figures below are reported *without* the minimum-region-size rule, because both are within-model comparisons at a fixed readout and the rule would shift every row by the same small amount.

on the same terms, and it is not a private advantage: applied at $f = 0.01$ it leaves every baseline row of Table 1 unchanged to three decimals except AKOrN (conv) on CLEVR foreground IoU, which improves from .4219 to .4267; the best-baseline row uses the better of the two for every baseline cell we measured.

No hyperparameter, threshold or rule of any model is selected on the 320 reserved images that Table 1 reports. Those images are read once, after every choice is frozen.

F Additional Experiments

F1 Budget and released weights Two conditions on Table 1 deserve measurements rather than argument: that the baselines are evaluated away from their published setting, and that the matched budget limits them.

Released weights. Only Slot Attention ships trained weights; the AKOrN, CAE and SynCx repositories release data but no checkpoints. Evaluated directly, at roughly $320\times$ our exposure budget, its released model scores .3747/.2219/.1165 on CLEVR, .4742/.2381/.2355 on Tetrominoes and .3205/.0980/.1204 on Multi-dSprites. Our model scores higher in all nine cells, by 0.377 to 0.720, so the comparison does not rest on our retraining of it. The extra data also helps it far more *off* its training distribution than on it — worth $+.0875$ FG-ARI on CLEVR but $+.3003$ on Tetrominoes and $+.1275$ on Multi-dSprites — so a data-efficiency conclusion drawn from an in-domain column alone would not transfer.

Budget. Comparing each baseline against its own published number, by its own evaluation code, isolates the effect of the shortened schedule. Two of the six are *not* budget-limited: AKOrN (conv) reaches 60.5 at $1/210$ of its schedule against a published 59.0 ± 4.3, and CAE reaches 10.3 at $1/64$ against a published 4.0. For those two the margin in Table 1 is not an artefact of a short schedule. The other four are limited — AKOrN (attn) 60.4 against 91.0 ± 0.5, ItrSA 58.1 against 82.5 ± 1.7, CAE++ 16.7 against 27.0, and CtCAE 12.0 against 54.0 — and against those the result should be read as a data-efficiency claim. We state both readings rather than only the favourable one. Full figures are in Appendix G.

F2 Rollout length and data scaling The readout integrates spiking activity over T simulation steps and reads the second half of the window. Table 2 varies T with the model fixed.

Halving the rollout costs 0.0019 FG-ARI — below any difference we can resolve at one seed — and *improves* foreground IoU. Performance degrades gently to $T = 256$ and collapses at $T = 128$, where the model also under-segments, predicting a foreground fraction of 0.1148 against a truth of 0.2115. We therefore use $T = 512$ as the default operating point and $T = 1024$ for the reported numbers. Shorter windows also fragment more: mean predicted regions per image rise from 8.0 at $T = 1024$ to 13.3 at $T = 256$, against 6.36 true objects.

Scaling with training data. Continuing training on additional unseen CLEVR images, in steps of 1,000 from 10,000 to 18,000, separates the three metrics. FG-ARI runs .8421, .8426, .8449, .8460, .8469, .8459, .8456, .8470, .8461: it gains .0028 over the first two steps and then stays inside a band of .0021 for the remaining six, with no trend. Foreground IoU runs .7545 to .7755 and matched-object IoU .6410 to .6579, both still rising at 18,000 and neither turning. More in-domain data therefore stops improving *how* patches are grouped at around 12,000 images while it is still improving *where the boundaries fall*. Mean predicted regions hold at 7.7–8.0 throughout, against 6.36 true objects, so the extra data sharpens the partition without changing how much it splits. The FG-ARI band is one seed wide and we read it as flat rather than converged; the two IoU curves have not been run far enough to find their own ceiling.

In-domain data is not the most useful data. The same 5,000 additional images buy more than twice as much when they come from a different dataset. Continuing on CLEVR raises CLEVR FG-ARI by $+.0038$; continuing on Tetrominoes instead raises it by $+.0084$, a $2.2\times$ larger gain at an identical budget. Together with the transfer result in Table 1, this argues that the model is not fitting a domain but learning a grouping rule that the extra renders refine.

F3 Oscillatory timing, measured The three measurements summarised in Section 3.2, in full. All run on the first 40 images of the CLEVR reserved split, on the same rollouts, so they are directly comparable. They are reported in the order they answer the hypothesis: which pattern is observed, whether its temporal alignment is necessary, and which signal carries it.

Validity of the ablation. The affinity Q is recomputed here from `component_spikes` rather than taken from the model, so that a modified spike trace can be fed through the same arithmetic. On the unshifted control the recomputed Q agrees with the model's own to 6.7×10^{-6} maximum absolute difference over all 4096^2 entries, so the ablations differ from the control in exactly one thing.

Separability. Ground-truth labels are used only to sort patch pairs into three groups and never to set a threshold or a count. Averaged over the 40 images, the affinity is 0.891 ± 0.111 for pairs inside one object, 0.047 ± 0.048 for pairs in different objects, and 0.099 ± 0.106 for object–background pairs. The within-object value is nineteen times the across-object value, and the across-object value sits near the floor of the measure.

	FG-ARI	FG-IOU	OBJ-IOU	regions
control	.8648	.8028	.6925	7.8
per-patch shift	.0735	.7358	.0325	562.3
gates, not spikes	.8648	.8028	.6925	7.8

Table 3: First 40 CLEVR reserved images. Destroying cross-patch timing collapses the partition (FG-ARI, OBJ-IOU) while barely moving foreground IoU. The whole readout, foreground decision included, is recomputed from the shifted traces; foreground IoU survives because it compares only the *union* of predicted non-background patches against the true foreground and is invariant to how that union is partitioned, and shattering one region into 562 leaves the union nearly unchanged. Substituting the phase gates for the spikes changes nothing at all.

Per-patch circular shift. Each patch's four component traces are rolled by one offset drawn uniformly over the readout window, independently per patch. Rate, waveform and within-patch structure survive; only the alignment between patches is destroyed.

The readout's *rules* are unchanged — the image-conditioned graph, the $Q \geq 0.5$ link threshold, the 8-patch component floor, the largest-component background heuristic and the greedy merge — but every quantity they operate on is recomputed from the shifted traces. In particular the foreground is *not* held fixed and is *not* rate-based: the readout passes every patch to the grouping step as a candidate and takes the largest surviving synchrony component as background, so the foreground decision is itself a function of the shifted affinity. The ablation therefore re-runs the whole readout and differs from the control in one input only.

Gates in place of spikes. The third row replaces `component_spikes` with `component_gates` and changes nothing else. Every mask is reproduced exactly, on all 40 images and to every digit of all three metrics. The reason is visible in the traces: with $s_{ia} = g_i e_{ia}$, the firing indicator e equals 1 in 99.992% of settled steps, the correlation between s and g is 0.9997, and the 0.008% of steps where the neuron is silent never change a thresholded decision. The timing that the shift ablation shows to be necessary is therefore the timing of the oscillatory gate. The spiking layer carries it without loss, but the readout cannot tell the two apart, and no claim should be made for the threshold nonlinearity on this evidence.

G Supplementary Tables

G1 Released weights These figures are given in prose in the main paper.

G2 Training budget per baseline

G3 Scaling with training data

	CLEVR	Tetrominoes	Multi-dSprites
Slot Attention, matched budget	.2872 / .2220 / .0801	.1739 / .2192 / .1071	.1930 / .0876 / .0680
Slot Attention, released weights	.3747 / .2219 / .1165	.4742 / .2381 / .2355	.3205 / .0980 / .1204
Ours, matched budget	**.8421 / .7545 / .6410**	**.8580 / .7885 / .8173**	**.6974 / .8177 / .7628**

Table 4: Released weights ($\approx$32M exposures) against the matched-budget retraining ($100{,}000$ exposures) and against our model. Triples are ARI / F-IOU / O-IOU. All three rows are reported *without* the minimum-region-size rule, so the released and retrained Slot Attention rows are compared on identical terms; Table 1 of the main paper applies the rule to every model and our row there reads $.8408/.7528/.6387$, $.8568/.7950/.8170$ and $.6966/.8190/.7625$.

Model	Budget	Ours	Published
AKOrN (conv)	$1/210$	**60.5**	59.0 ± 4.3
CAE	$1/64$	**10.3**	4.0
AKOrN (attn)	$1/210$	60.4	91.0 ± 0.5
ItrSA	$1/210$	58.1	82.5 ± 1.7
CAE++	$1/64$	16.7	27.0
CtCAE	$1/64$	12.0	54.0

Table 5: Each baseline at our budget against its own published figure, by its own metric and evaluation code. *Budget* is the fraction of the published training schedule our matched-exposure protocol gives it. The top block is *not* budget-limited: AKOrN (conv) exceeds its published CLEVR number at $1/210$ of its schedule and CAE exceeds its own at $1/64$. The bottom block is.

Images	FG-ARI	FG-IOU	OBJ-IOU	Δ FG-ARI
10,000	.8421	.7545	.6410	—
11,000	.8426	.7539	.6427	$+.0005$
12,000	**.8449**	**.7607**	**.6461**	$+.0028$

Table 6: *Images* is the cumulative count of unique training images. Each row adds $1{,}000$ previously unseen CLEVR images for 10 epochs, resuming from the previous row. CLEVR evaluation.

Symbol	Shape	Meaning and status
γ	$N \times 8$	Computed patch activations
W	16×8	Trainable graph projection
$\boldsymbol{z}_i$	16	Computed normalized patch embedding
S, d, c	$N \times N$ each	Computed scores, geodesic distances, and local costs
τ, r, G	Scalar each	Positive values derived from trainable scalars
A, B	$N \times N$ each	Computed connection matrices
W_{drive}, W_ϕ	4×8; scalar	Trainable input projection and phase gain
$\boldsymbol{u}, \bar{\boldsymbol{u}}, \boldsymbol{\phi}$	$N \times 4$ each	Computed input representations

Table 7: Variable meanings, shapes, and parameter status for the sensory representation.

Symbol	Shape	Meaning and status
$\boldsymbol{\theta}(t), \boldsymbol{\omega}, \boldsymbol{\kappa}$	$N \times 4$ each	Dynamic phase; trainable natural frequencies and drive strengths
f	Scalar	Trainable frequency gain
$K, \Delta t$	Scalar each	Fixed coupling scale and time step
R	$N \times N$	Trainable directional matrix
$\bar{A}, C, \bar{C}, \alpha$	$N \times N$ each	Computed corrected graph, inverse costs, normalized costs, and lags
$\bar{\theta}(t), g(t)$	N each	Computed mean phases and gates
$\boldsymbol{x}(t)$	$N \times 4$	Computed gated features
$U, \boldsymbol{b}$	4×2; 4	Trainable branch projection and bias
$\boldsymbol{v}_{ia}(t), \boldsymbol{h}_{ia}(t), \widehat{\boldsymbol{h}}_{ia}(t)$	4 each	Computed branch input and states
$\boldsymbol{k}, \boldsymbol{\beta}$	$N \times 4$ each	Trainable retention parameters and their sigmoid transforms
η, ξ	4 each	Trainable dendritic and membrane modulation weights
$H(t), V(t), \widehat{V}(t), e(t), s(t)$	$N \times 4$ each	Computed membrane input, voltages, events, and spikes
ℓ, λ	N each	Trainable retention parameters and their sigmoid transforms
q, ϑ	4 each	Trainable threshold parameters and their derived thresholds

Table 8: Variable meanings, shapes, and parameter status for the recurrent core.

Symbol	Shape	Meaning and status
$\boldsymbol{r}_{ia}$	512	Computed normalized temporal trace
ρ	$N \times N \times 4$	Computed component correlations
Q, Q', E	$N \times N$ each	Computed affinity, stabilized affinity, and signed evidence
Δ	Scalar per pair	Computed merge score
y	N or 64×64	Computed hard labels
M_k	N or 64×64	Computed binary mask for object group k

Table 9: Variable meanings, shapes, and parameter status for the mask readout.